

8/3_Python_Day1_종합실습

- P037_김규현

1. 환경 구성

```
python3.11 -m venv .venv
```

- requirements.txt

```
httpx
pydantic>=2
pandas
pyarrow
pytest
ruff
```

2. 비동기 수집

- 3개 API 모두 asyncio.gather()로 동시 수집, 응답 정상 확인

```
All checks passed! /Users/gh/SKALA/... Python/day1_practice/.venv/bin/python /Users/gh/SKALA/... Python/day1_practic
e/src/collectors.py
수집 완료
수집 완료
수집 완료
{'latitude': 37.55, 'longitude': 127.0, 'generationtime_ms': 0.056743621826171875, 'utc_offset_seconds': 32400, 'timezone': 'Asia/Seoul', 'timezone_abbreviatio
n': 'GMT+9', 'elevation': 34.0, 'hourly_units': {'time': 'iso8601', 'temperature_2m': 'C', 'precipitation_probability': '%', 'hourly': {'time': ['2026-08-03T
00:00', '2026-08-03T01:00', '2026-08-03T02:00', '2026-08-03T03:00', '2026-08-03T04:00', '2026-08-03T05:00', '2026-08-03T06:00', '2026-08-03T07:00', '2026-08-03
T08:00', '2026-08-03T09:00', '2026-08-03T10:00', '2026-08-03T11:00', '2026-08-03T12:00', '2026-08-03T13:00', '2026-08-03T14:00', '2026-08-03T15:00', '2026-08-0
3T16:00', '2026-08-03T17:00', '2026-08-03T18:00', '2026-08-03T19:00', '2026-08-03T20:00', '2026-08-03T21:00', '2026-08-03T22:00', '2026-08-03T23:00', '2026-08-0
4T00:00', '2026-08-04T01:00', '2026-08-04T02:00', '2026-08-04T03:00', '2026-08-04T04:00', '2026-08-04T05:00', '2026-08-04T06:00', '2026-08-04T07:00', '2026-08-0
4T08:00', '2026-08-04T09:00', '2026-08-04T10:00', '2026-08-04T11:00', '2026-08-04T12:00', '2026-08-04T13:00', '2026-08-04T14:00', '2026-08-04T15:00', '2026-0
8-04T16:00', '2026-08-04T17:00', '2026-08-04T18:00', '2026-08-04T19:00', '2026-08-04T20:00', '2026-08-04T21:00', '2026-08-04T22:00', '2026-08-04T23:00', '2026-0
8-05T00:00', '2026-08-05T01:00', '2026-08-05T02:00', '2026-08-05T03:00', '2026-08-05T04:00', '2026-08-05T05:00', '2026-08-05T06:00', '2026-08-05T07:00', '2026-0
8-05T08:00', '2026-08-05T09:00', '2026-08-05T10:00', '2026-08-05T11:00', '2026-08-05T12:00', '2026-08-05T13:00', '2026-08-05T14:00', '2026-08-05T15:00', '2026-08-05T16:00', '2026-08-05T17:00', '2026-08-05T18:00', '2026-08-05T19:00', '2026-08-05T20:00', '2026-08-05T21:00', '2026-08-05T22:00', '2026-08-05T23:00'}], 't
emperature_2m': [25.3, 24.9, 24.5, 24.2, 24.0, 23.7, 23.6, 24.2, 24.5, 25.3, 28.1, 30.8, 32.8, 34.0, 34.7, 35.0, 35.0, 34.7, 33.5, 31.8, 30.5, 29.6, 28.9, 28.4,
27.9, 27.7, 27.7, 27.4, 27.0, 26.8, 26.6, 26.9, 27.9, 29.2, 30.5, 31.9, 33.0, 34.0, 34.8, 35.3, 35.4, 35.1, 34.5, 33.4, 32.1, 31.1, 30.2, 29.6, 29.0, 28.4, 2
7.8, 27.4, 26.4, 26.0, 25.6, 25.9, 26.9, 28.3, 29.9, 31.2, 32.4, 33.2, 33.8, 34.4, 34.8, 34.9, 34.4, 32.8, 31.4, 30.3, 29.1, 28.1], 'precipitation_probability'
: [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 4, 6, 8, 10, 10, 7, 3, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 2, 6, 14, 24, 37, 59, 83, 92, 73, 38, 12, 5, 6, 8
, 10, 14, 16, 15, 13, 10, 7, 3, 0, 0, 1, 2, 3, 5, 6, 7, 7, 6, 4, 2, 0, 0, 1]}
{'area': 100210, 'cioc': 'KOR', 'flag': 'KR', 'gini': 31.4, 'name': 'Korea (Republic of)', 'flags': {'png': 'https://flagcdn.com/w320/kr.png', 'svg': 'https://
flagcdn.com/kr.svg'}, 'latlng': [37, 127.5], 'region': 'Asia', 'borders': ['PRK'], 'capital': 'Seoul', 'demonym': 'South Korean', 'languages': [{'name': 'Korea
n', 'iso639_1': 'ko', 'iso639_2': 'kor', 'nativeName': '한국어'}], 'subregion': 'Eastern Asia', 'timezones': ['UTC+09:00'], 'alpha2Code': 'KR', 'alpha3Code': '
KOR', 'currencies': [{'code': 'KRW', 'name': 'South Korean won', 'symbol': '₩'}], 'nativeName': '대한민국', 'population': 51780579, 'independent': True, 'numeri
cCode': '410', 'callingCodes': ['82'], 'topLevelDomain': ['.kr'], 'populationDensity': 516.72}
{'status': 'success', 'country': 'United States', 'countryCode': 'US', 'region': 'VA', 'regionName': 'Virginia', 'city': 'Ashburn', 'zip': '20149', 'lat': 39.0
3, 'lon': -77.5, 'timezone': 'America/New_York', 'isp': 'Google LLC', 'org': 'Google Public DNS', 'as': 'AS15169 Google LLC', 'query': '8.8.8.8'}
```

3. 비동기 수집 → Pydantic 검증 → csv,parquet 저장 및 쓰 기 측정 비교

```
e/src/main.py
3개 API 데이터 수집을 시작합니다.
수집 완료
수집 완료
수집 완료
모든 API 데이터가 Pydantic 검증을 통과했습니다.

[weather] 저장 성능 비교
CSV 쓰기: 0.003587초, 읽기: 0.003851초
Parquet 쓰기: 0.030207초, 읽기: 0.036986초

[country] 저장 성능 비교
CSV 쓰기: 0.000462초, 읽기: 0.000284초
Parquet 쓰기: 0.000441초, 읽기: 0.000523초

[ip] 저장 성능 비교
CSV 쓰기: 0.000244초, 읽기: 0.000253초
Parquet 쓰기: 0.000419초, 읽기: 0.000494초

모든 데이터를 data 폴더에 저장했습니다.
```

→ 작은 데이터 양에서는 parquet의 초기화 비용 때문에 CSV가 더 빠르게 측정될 수 있음

4. pytest 및 ruff check

- `pyproject.toml`

```
[tool.pytest.ini_options]
pythonpath = [".", "src"]
testpaths = ["tests"]

[tool.ruff]
line-length = 88
target-version = "py311"

[tool.ruff.lint]
select = ["E", "F", "I"]
```

```
(.venv) → day1_practice git:(main) ✗ pytest -q
..... [100%]
8 passed in 0.43s
(.venv) → day1_practice git:(main) ✗ ruff check .
All checks passed!
```

5. Git commit

```

(.venv) → 실습파일 git:(main) x git add .gitignore Python/
(.venv) → 실습파일 git:(main) x git commit -m "Add Python practice file"
[main d4b4065] Add Python practice file
34 files changed, 1372 insertions(+)
create mode 100644 Python/Day1/Practice1.py
create mode 100644 Python/Day1/Python_Practice1_Data.json
create mode 100644 Python/Day1/Python_Practice2_Data.json
create mode 100644 Python/Day1/error_sales.json
create mode 100644 Python/Day1/example.py
create mode 100644 Python/Day1/ppp.py
create mode 100644 Python/Day1/test.py
create mode 100644 Python/Day1/valid_sales.csv
create mode 100644 "Python/Day1/\355\214\220\352\265\220_2\353\260\230_\352\271\200\352\267\234\355\230\204.py"
create mode 100644 "Python/Day1/\355\214\220\352\265\220_2\353\260\230_\352\271\200\352\267\234\355\230\204_practice1.py"
create mode 100644 Python/day1_practice/README.md
create mode 100644 Python/day1_practice/data/country.csv
create mode 100644 Python/day1_practice/data/country.parquet
create mode 100644 Python/day1_practice/data/ip.csv
create mode 100644 Python/day1_practice/data/ip.parquet
create mode 100644 Python/day1_practice/data/weather.csv
create mode 100644 Python/day1_practice/data/weather.parquet
create mode 100644 Python/day1_practice/pyproject.toml
create mode 100644 Python/day1_practice/requirements.txt
create mode 100644 Python/day1_practice/src/__init__.py
create mode 100644 Python/day1_practice/src/__pycache__/__init__.cpython-311.pyc
create mode 100644 Python/day1_practice/src/__pycache__/collectors.cpython-311.pyc
create mode 100644 Python/day1_practice/src/__pycache__/main.cpython-311.pyc
create mode 100644 Python/day1_practice/src/__pycache__/models.cpython-311.pyc
create mode 100644 Python/day1_practice/src/__pycache__/storage.cpython-311.pyc
create mode 100644 Python/day1_practice/src/__pycache__/validators.cpython-311.pyc
create mode 100644 Python/day1_practice/src/collectors.py
create mode 100644 Python/day1_practice/src/main.py
create mode 100644 Python/day1_practice/src/models.py
create mode 100644 Python/day1_practice/src/storage.py
create mode 100644 Python/day1_practice/src/validators.py
create mode 100644 Python/day1_practice/tests/__pycache__/test_models.cpython-311-pytest-9.1.1.pyc
create mode 100644 Python/day1_practice/tests/test_models.py
(.venv) → 실습파일 git:(main) x git push origin main

```